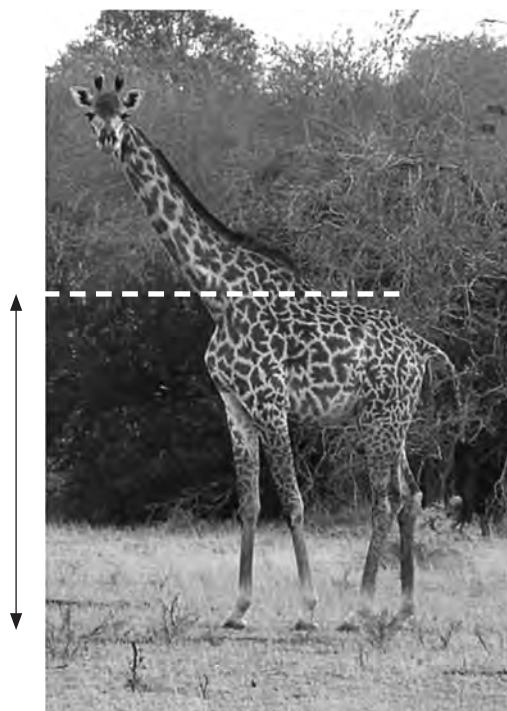


- 7 The photograph shows an adult giraffe.

height at shoulders = 3.0 m



- (a) Estimate the length of the neck of this giraffe from the shoulders, using the information in the photograph.

(1)

Answer m

- (b) The heart of an adult giraffe can be 60 cm long.

Explain why the heart of a giraffe needs to be so large.

(2)

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(c) The arteries near the heart of a giraffe are highly elastic.

(i) Draw a labelled diagram of an artery, as seen in section.

(3)

(ii) Explain why the arteries near the heart of a giraffe are highly elastic.

(2)

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P 6 1 4 6 9 A 0 2 3 2 8

(d) Damage to the legs of the giraffe could result in excessive bleeding.

To prevent excessive bleeding, the capillaries near the surface of the skin are very narrow.

(i) Explain why very narrow capillaries prevent excessive bleeding.

(2)

(ii) The red blood cells of the giraffe are about one third the size of human red blood cells, so that they can pass through the very narrow capillaries.

The small size of the red blood cells ensures that the legs of the giraffe have a good supply of oxygen.

Explain why smaller red blood cells increase the supply of oxygen to the legs.

(2)

(Total for Question 7 = 12 marks)



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